3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
Row 3 Column 2 = Row 4 Column 2 = 0
Row 4 Column 4 = 0
$\therefore$ The condition for each column having a ~~pi~~ leading entry, all the positions below the leading entry must contain only 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 4 = Row 2 Column 4 = 0
Row 1 Column 2 = 0
$\therefore$ The condition for each column having a leading entry, all the positions above the leading entry must contain only 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form $\square$

Only the first 3 rows contain a pivot position.
Since B does not have a pivot position in every row, By Theorem 4, the columns of B does not span R^4 and ~~Ax=b does not have a solution for each b in R^4~~ $Bx = y$ does not have a solution for each y in R^4. $\square$

19) Can each vector in R^4 be written as a linear combination of the columns of the matrix A above? Do the columns of A span R^4?

$\boxed{\text{Solution:}}$ From Problem 17, we have seen that equation $Ax = b$ does not have a solution for each b in R^4. By Theorem 4, each vector in R^4 cannot be written as a linear combination of the columns of the matrix A. Also, the columns of A do not span R^4 from Theorem 4. $\square$

20) Can every vector in R^4 be written as a linear combination of the columns of the matrix B above? Do the columns of B span R^3? ?? ??

$\boxed{\text{Solution:}}$ From Problem 18, we know the equation $Bx = y$ does not have a solution for each y in R^4. $\therefore$ By Theorem 4, every vector in R^4 cannot be written as a linear combination of the columns of the matrix B.
(Second part I don't know)